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DEPARTMENT OF INFORMATION TECHNOLOGY



NM1020 – UI&UX DESIGN(NAAN MUDALVAN)

Name :

Register Number :

Year & Branch :

Semester :

Academic Year :

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CERTIFICATE

This is to Certify that the Bonafide record of the practical work done by
..... Register Number :
..... of III year **B.TECH (INFORMATION
TECHNOLOGY)** submitted for the **B.TECH IT** practical examination (V
Semester) in **NM1020 – UI&UX DESIGN(NAAN MUDALVAN)** during the
academic year 2025 – 2026 .

Staff in-Charge

Head of the Department

Submitted for the practical examination held on _____ .

Internal Examiner

External Examiner

Introduction to UI & UX Design

(Naan Mudhalvan – NM1020)

1. What is UI & UX Design?

UI (User Interface) and **UX (User Experience)** are two key parts of digital product design that work hand-in-hand to create user-friendly, attractive, and effective applications or websites.

- **User Interface (UI)** focuses on the **visual design and layout** — how things look on the screen. It includes buttons, icons, color schemes, typography, spacing, and imagery.
- **User Experience (UX)** focuses on the **overall journey of the user** — how easy, efficient, and enjoyable it is to use the product. It involves research, user flows, wireframes, and usability testing.

In simple terms:

UI is what people see — UX is what people feel.

A good UI/UX design ensures that the product is **visually appealing**, **intuitive**, and **helps users achieve their goals quickly**.

2. Difference Between UI and UX

Aspect	UI (User Interface)	UX (User Experience)
Focus	Looks and aesthetics	Feel and functionality
Goal	To make the interface attractive and consistent	To make the experience smooth and satisfying
Includes	Colors, fonts, layouts, icons, visual styles	User research, personas, wireframes, user testing

Aspect	UI (User Interface)	UX (User Experience)
Example	The button color, shape, and animation	How easy it is to complete a task using that button

3. Importance of UI/UX Design

- Helps users navigate easily
- Builds trust and brand identity
- Reduces user frustration and errors
- Increases engagement and conversion rate
- Improves accessibility and usability for all

A well-designed interface creates a **positive impression**, while good user experience keeps users **coming back**.

What is Figma?

Figma is a **cloud-based UI/UX design tool** used for creating digital designs, prototypes, and design systems collaboratively. It allows multiple designers to work together in real-time — similar to Google Docs, but for design.

Key Features of Figma:

1. **Web-Based Collaboration** – No installation required; works directly in the browser.
2. **Vector Editing** – Supports precise and scalable vector-based design.
3. **Components and Variants** – Reusable design elements for consistency.
4. **Auto Layout** – Automatically adjusts spacing and alignment when content changes.

5. **Prototyping** – Create interactive flows and animations without writing code.
 6. **Design Systems** – Build and maintain brand guidelines in one shared library.
 7. **Developer Handoff** – Developers can easily inspect, copy, and export assets.
-

4. Why Figma is Used in UI/UX Design

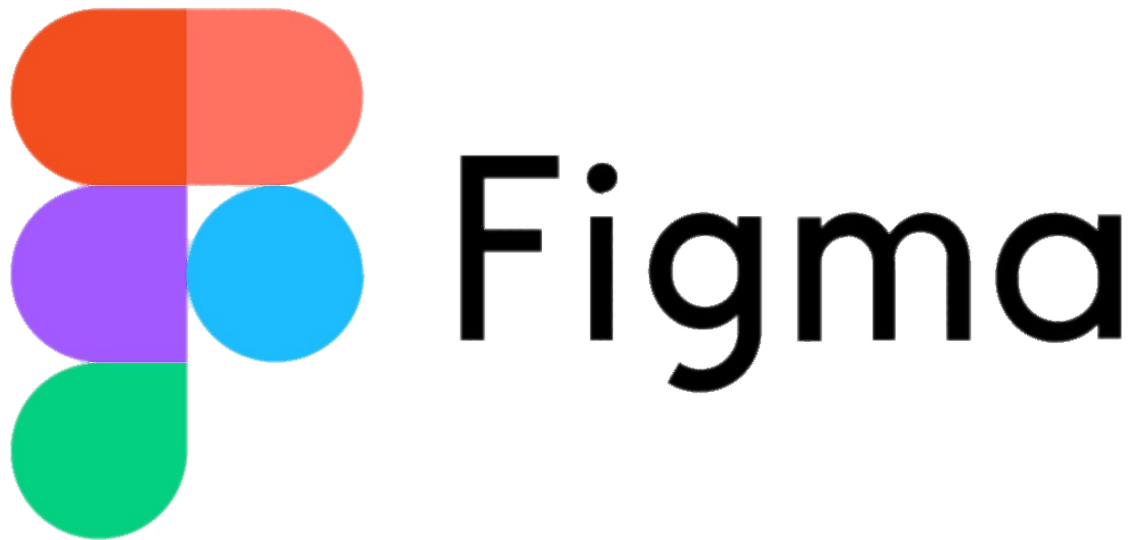
- Enables **collaboration** between designers, developers, and clients.
 - Speeds up the **design-to-prototype** process.
 - Supports **responsive design** for multiple devices.
 - Helps create **wireframes, mockups, and prototypes** in one tool.
 - Cloud storage makes it easy to access and update projects anywhere.
-

5. Figma in Naan Mudhalvan Course

In the **Naan Mudhalvan (NM1020)** course, Figma is used to:

- Learn **practical UI/UX design** skills.
- Build **mini projects** like task management, e-commerce, or health tracking apps.
- Understand how to design for **both mobile and web platforms**.

- Showcase your design process — from **user research** to **final prototype** — in one place.
-



6. Conclusion

UI and UX design together form the **heart of digital product creation**. With tools like **Figma**, students can practically explore user-centered design — focusing not just on how an app looks, but on how it works for real users.

By mastering Figma and UX principles, designers can create experiences that are both **beautiful and meaningful**, improving lives one interface at a time.

Naan Mudhalvan – UI/UX Design using Figma

Title: TaskEase – A Simple Task Management Application

Aim:

To design and prototype a clean, minimal, and user-friendly **Task Management Application** using **Figma**, applying UI/UX principles to enhance user experience and productivity.

Tools Required:

- **Figma** – For interface design and prototyping
- **Google Docs / Sheets** – For documentation and feedback
- **Icons8 / Material Icons** – For icons and UI assets

Objective:

The primary objective of the **TaskEase project** is to design a **digital platform** that allows users to efficiently manage their daily tasks with **minimal effort and maximum clarity**. The project seeks to establish a **seamless connection between usability and visual design**, addressing real-world productivity challenges through a user-centered approach.

Key Objectives:

1. **To identify and analyze pain points** of existing task management applications through **user research and surveys**.
2. **To design a clean and distraction-free interface** that prioritizes essential features such as **task creation, categorization, and progress tracking**.
3. **To implement an information architecture** that ensures smooth navigation and minimal cognitive load for users.
4. **To build an interactive high-fidelity prototype using Figma** that accurately represents user flows, actions, and interactions.
5. **To validate the usability and effectiveness of the design** through user testing, feedback collection, and iterative improvements.
6. **To enhance accessibility and ensure adherence to modern UI/UX design standards**, including **responsive layouts, optimal color contrast, and touch-friendly components**.

The overarching goal is to **transform user research insights into a human-centered digital experience** that promotes **clarity, efficiency, and daily productivity**.

Problem Statement:

Existing task management apps are often cluttered or complex. Users struggle with navigation and focus. The goal is to create a **lightweight,**

simple, and efficient task app that helps users stay organized without distractions.

Abstract

The **TaskEase project** focuses on the design and development of a **user-centric task management application** that simplifies the process of organizing and monitoring daily activities. In the modern digital environment, individuals rely heavily on productivity tools to manage their personal and professional tasks; however, existing solutions often suffer from **complexity, excessive features, and cluttered interfaces**.

The **TaskEase application** aims to overcome these limitations by providing a **minimalistic, intuitive, and aesthetically pleasing user interface** that enhances user productivity while maintaining focus and simplicity. Built using **Figma**, the project emphasizes key **User Experience (UX)** principles such as **accessibility, usability, and cognitive load reduction**.

Through **user research, iterative wireframing, and high-fidelity prototyping**, the application demonstrates a practical implementation of **UI/UX design thinking** in real-world productivity scenarios. The final prototype showcases an **efficient workflow, responsive design, and user-friendly navigation** that collectively promote **better task organization and user satisfaction**.

User Research:

- **Target Users:** Students and working professionals.
- **User Goals:**
 - Add and view daily tasks quickly

- Set priorities and deadlines

- Track progress easily

- **User Pain Points:**

- Too many features in existing apps

- Poor visual organization

- No clear task tracking progress

User Persona:

A representation of a typical user to guide design decisions.

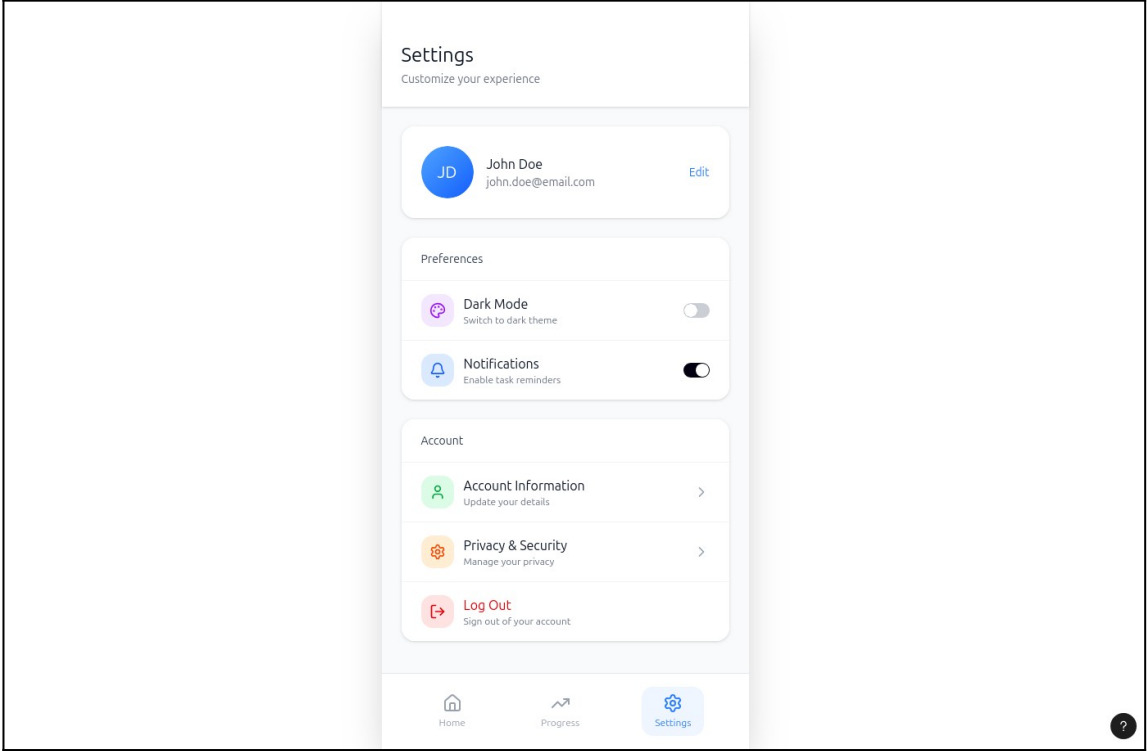
Name: Priya

Age: 21

Occupation: College Student

Goals: Wants a simple way to plan her assignments and personal tasks.

Frustrations: Finds most apps confusing and filled with unnecessary features.



Procedure

The development of the **TaskEase application** followed a **structured UI/UX design process** based on the **principles of Design Thinking**. Each stage was executed **iteratively** to ensure that **user needs, feedback, and functionality** were continuously incorporated into the design.

1. Requirement Gathering and User Research

The process began by identifying the **target audience** — students, working professionals, and individuals seeking a simple productivity solution.

Surveys and informal interviews were conducted to understand their **daily workflows, challenges, and expectations** from a task management app.

The collected data was analyzed to reveal common issues such as **complex navigation, visual clutter, and lack of focus-oriented features** in existing applications.

2. Problem Definition and Ideation

From the research insights, the design problem was clearly defined:

Users need a **lightweight, distraction-free, and intuitive interface** to manage daily tasks efficiently.

Brainstorming sessions were conducted to generate ideas for layout, task categorization, and interaction flow.

Concepts were then **prioritized based on simplicity, usability, and relevance** to user needs.

3. Information Architecture and Wireframing

A **hierarchical structure** was developed to define how users navigate between screens and actions.

Low-fidelity **wireframes** were created in **Figma** to visualize layout and content organization.

This step helped test the **navigation flow and usability** before proceeding to detailed UI design.

4. UI Design and Prototyping in Figma

High-fidelity **prototypes** were designed in **Figma**, applying consistent **color palettes, typography, iconography, and spacing** based on the mood board and design system.

Interactive **prototype links** were created to simulate real-world user journeys and demonstrate app functionality dynamically.

5. Usability Testing and Feedback Analysis

The prototype was tested by a small group of users to evaluate **ease of use, clarity, and responsiveness**.

Feedback led to refinements such as improved **task visibility**, smoother transitions, and the addition of **dark mode** for accessibility.

This ensured a design that was both **user-friendly and adaptive**.

6. Documentation and Final Review

The final design was **documented and annotated** with explanations for each design decision, supported by **user research data and usability outcomes**.

This comprehensive documentation demonstrated how **Design Thinking principles** were effectively used to solve real-world interaction and productivity challenges.

Information Architecture:

The **Information Architecture (IA)** of *TaskEase* was designed to ensure a **seamless and intuitive user journey** from entry to task completion. The app flow focuses on **clarity, minimal steps, and logical grouping** of related actions to reduce cognitive load and improve usability.

Overall Structure:

1. **Splash Screen** – Displays the application logo and a brief welcome message, setting the visual tone and brand identity.
2. **Home Screen** – Serves as the central hub where all tasks are displayed, organized by **priority, deadline, or status** (completed/incomplete).
3. **Add Task Screen** – Enables users to create new tasks with options to **set deadlines, assign priorities, and add notes or descriptions**.
4. **Progress Screen** – Provides visual insights into user productivity using **charts, bars, and percentage indicators** representing completed tasks.
5. **Settings Screen** – Allows customization through **theme selection (light/dark), notification preferences, and profile settings**.

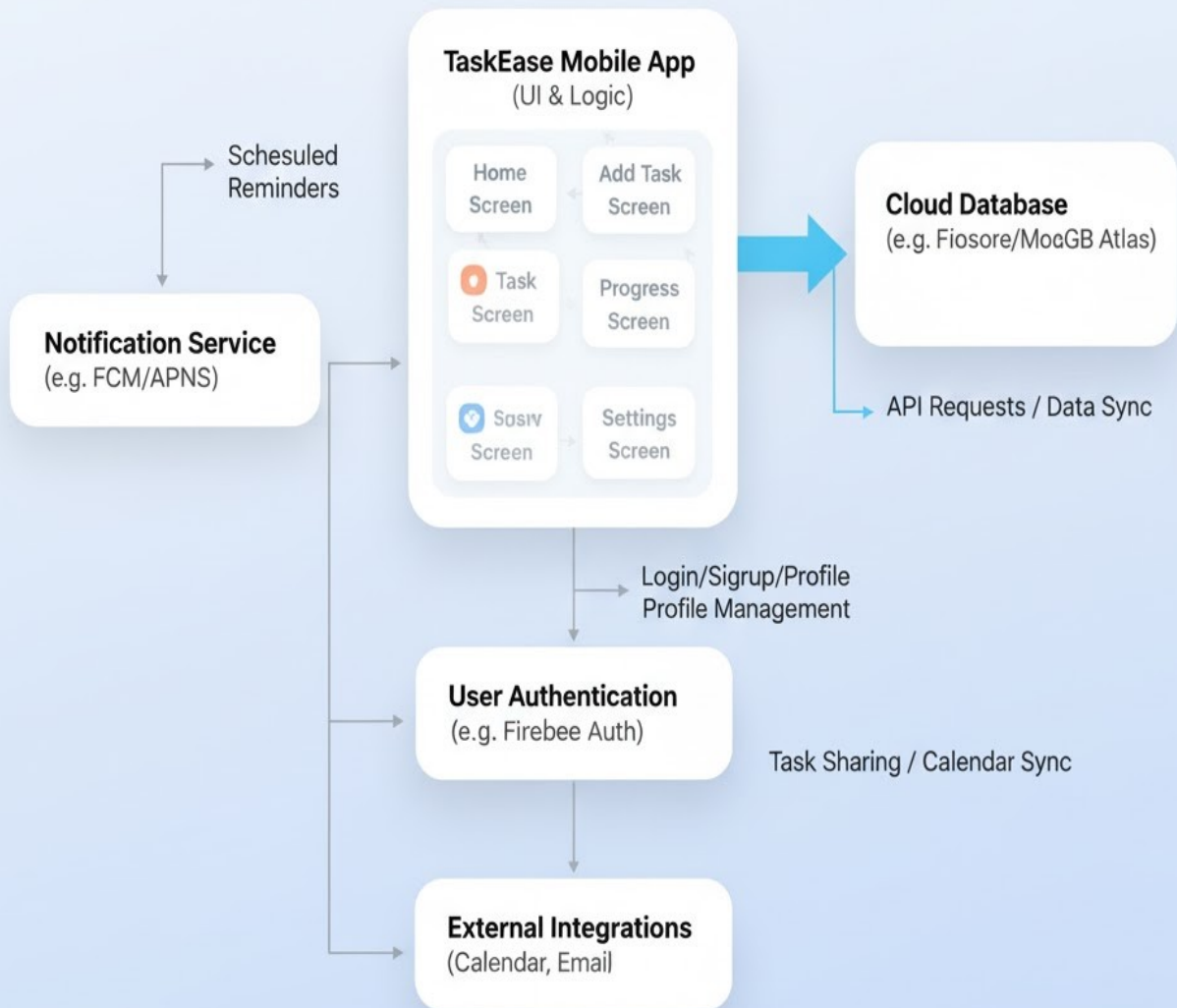
Navigation Design:

The navigation follows a **flat hierarchy**, allowing users to access any major feature within **two taps**. This structure minimizes user effort and promotes efficiency.

Consistent **icons, color codes, and typography** are used throughout the interface to enhance the recognition of actions and categories.

Each screen connects logically to the next, ensuring **consistency, predictability, and flow** — which are core elements of an effective **User Experience (UX)** design.

TaskEase App Workflow Architecture



Wireframe Design :

Wireframing served as the **foundation** for visualizing the **layout, functionality, and flow** of the *TaskEase* application before developing the high-fidelity prototype. The **low-fidelity wireframes** were created using **Figma** to define structure, spacing, and content organization **without visual distractions**, focusing purely on usability and information flow.

Purpose of Wireframes:

1. To outline the **skeletal framework** of the interface and its core components.
2. To identify and test **navigation paths** early in the design process.
3. To ensure that **user goals are achievable** in minimal steps with intuitive layouts.

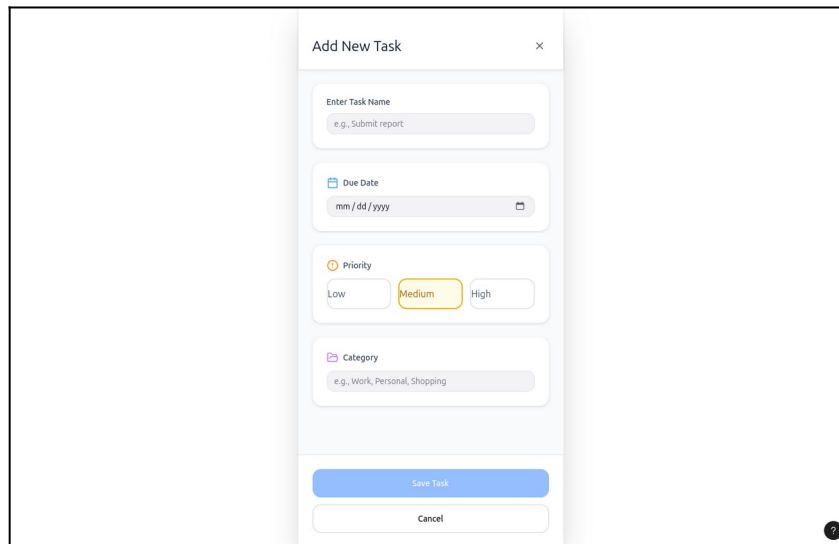
Key Wireframe Elements:

- **Home Screen:** Features a vertical list layout displaying **task categories, due dates, and checkboxes** for task completion.
- **Add Task Screen:** Contains **input fields** for title, description, and date selection, along with **priority icons** for quick categorization.
- **Progress Screen:** Displays **bar graphs or circular progress charts** to represent completion rates and productivity levels.
- **Settings Screen:** Includes simple **toggle buttons** for managing **notifications, themes, and account preferences**.

The wireframes were validated through **feedback sessions and peer reviews**, leading to refinements in **spacing, visual hierarchy, and**

button placement.

This **iterative process** ensured that the final design not only looked visually appealing but also met the **functional and experiential needs** of users.



UI Design (High-Fidelity Prototype):

Designed using **Figma**, focusing on clarity, spacing, and color harmony.

Color Palette: Soft blues, white, and gray tones

Font Used: Poppins / Inter

Style: Minimal, modern, and consistent



TaskEase

Plan. Prioritize. Progress.

Today's Tasks

- ☒ Submit report
 - ☐ Buy groceries
 - ☐ Schedule dentist
 - ☐ Schedule dentist
 - ☐ Category
 - ☐ Notes
- [Save](#) [Cancel](#)



Home



Progress



Settings

Enter Task Name



You're doing great!



Completed Tasks

5

Pending Tasks

2

☒ Submit report

☐ Read book

☐ Buy groceries

☐ Plan next week



John Doe



Theme

Light/Dark mode



Notifications



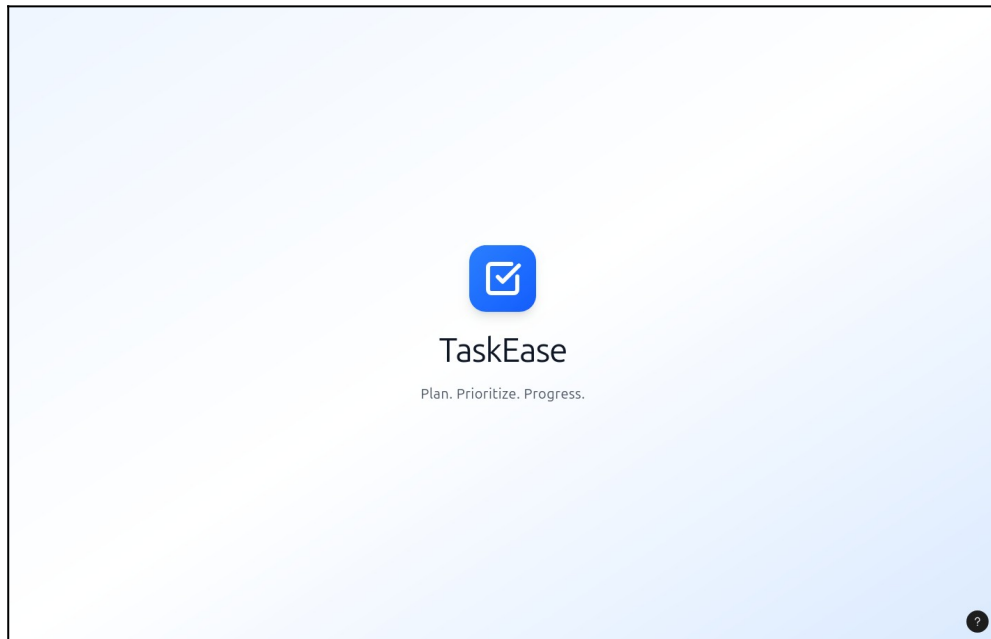
Account Info

[Log Out](#)

Screens Designed:

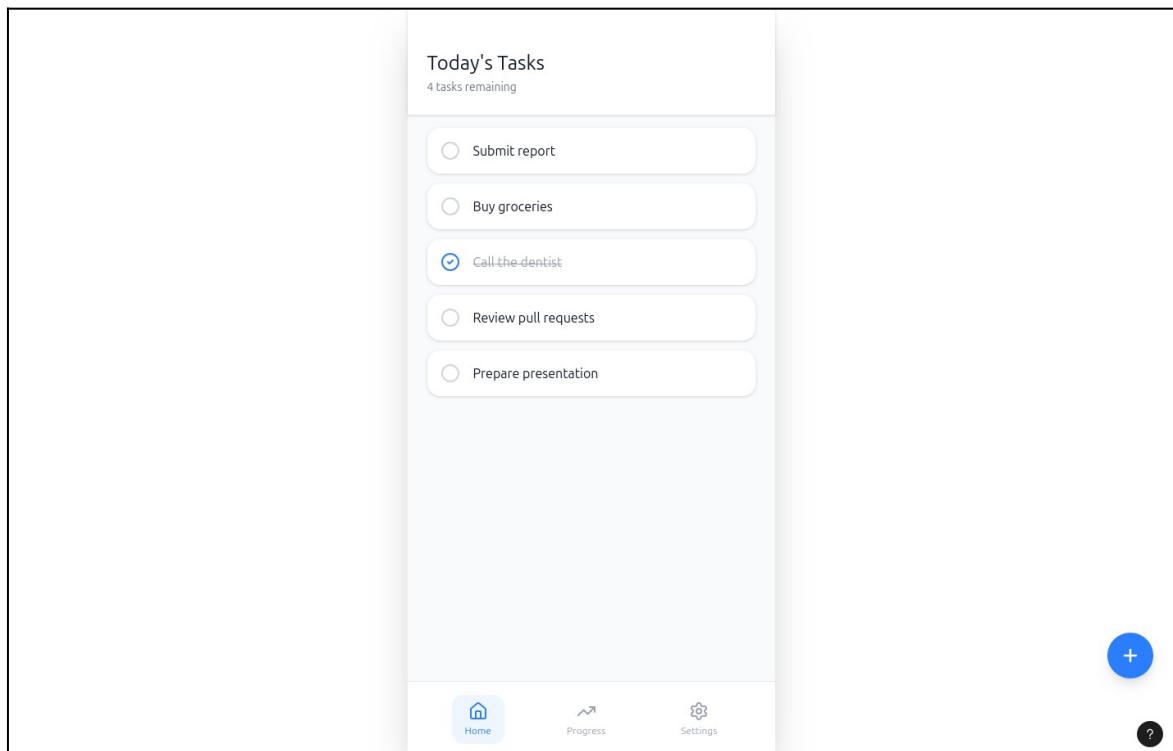
1. *Splash Screen*

App logo and welcome message.



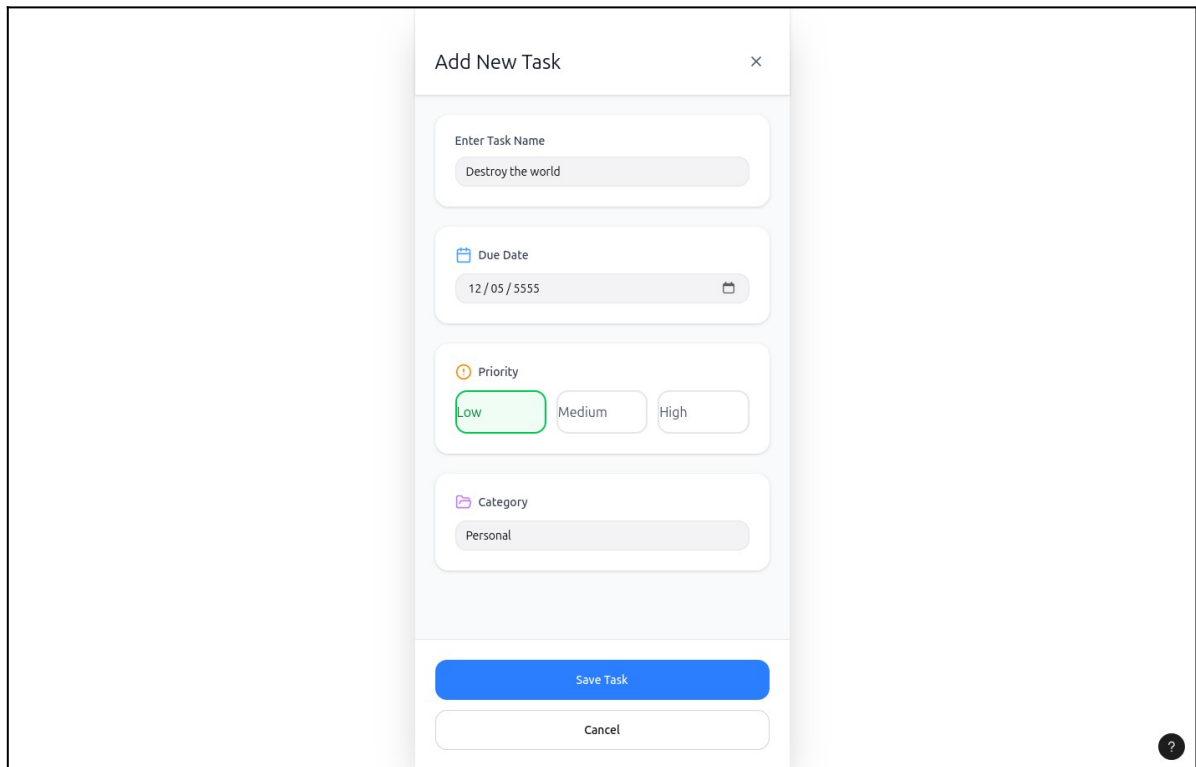
2. *Home Screen*

Displays a list of daily tasks with checkboxes and categories.



3. Add Task Screen

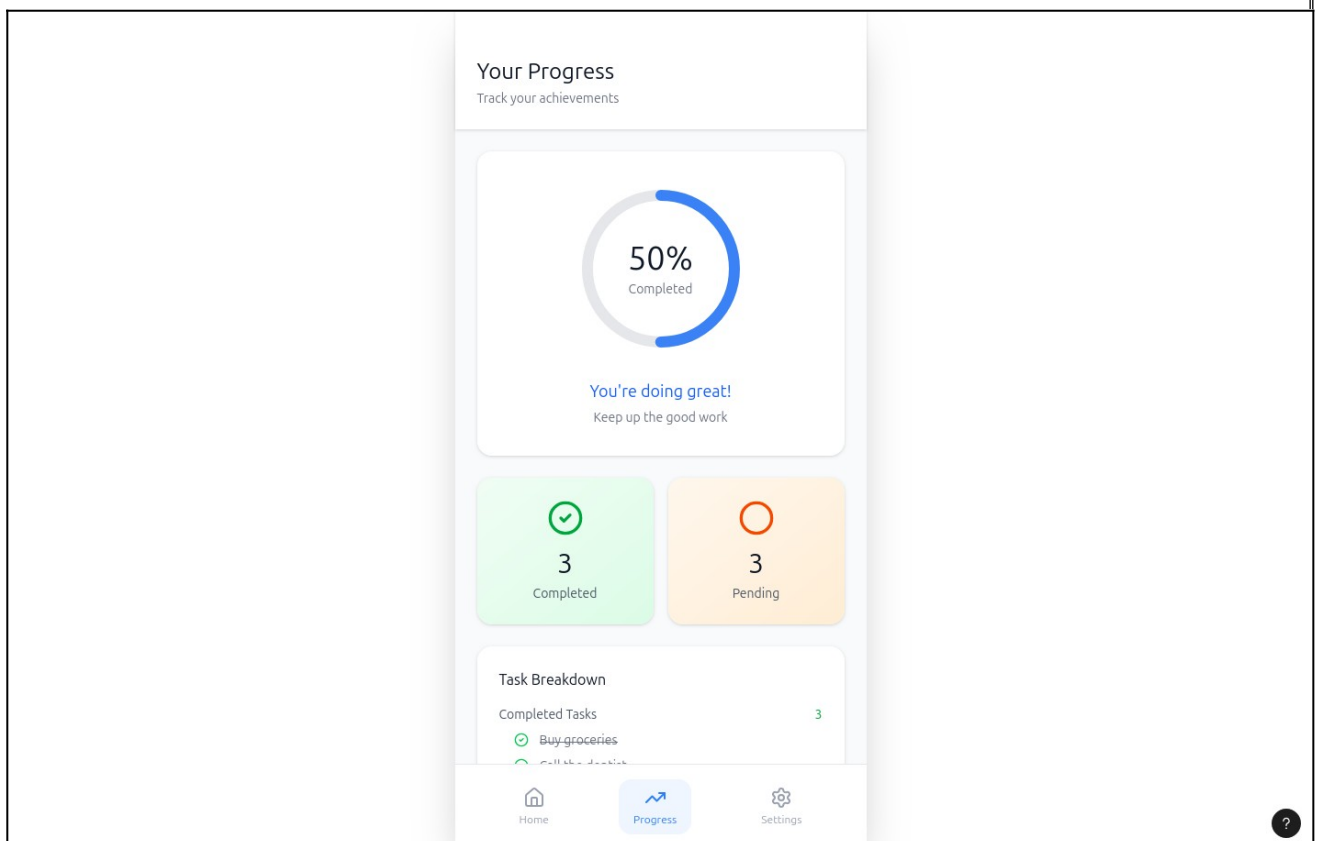
Allows users to create new tasks, select priorities, and set due dates.



The 'Add New Task' screen is a mobile app interface for creating tasks. It features a title bar 'Add New Task' with a close button. Below the title bar are four input fields: 'Enter Task Name' (with the placeholder text 'Destroy the world'), 'Due Date' (with the placeholder text '12/05/5555' and a calendar icon), 'Priority' (with three radio buttons: 'Low' (selected), 'Medium', and 'High'), and 'Category' (with the placeholder text 'Personal'). At the bottom are two buttons: 'Save Task' (blue) and 'Cancel' (white). A small question mark icon is located in the bottom right corner.

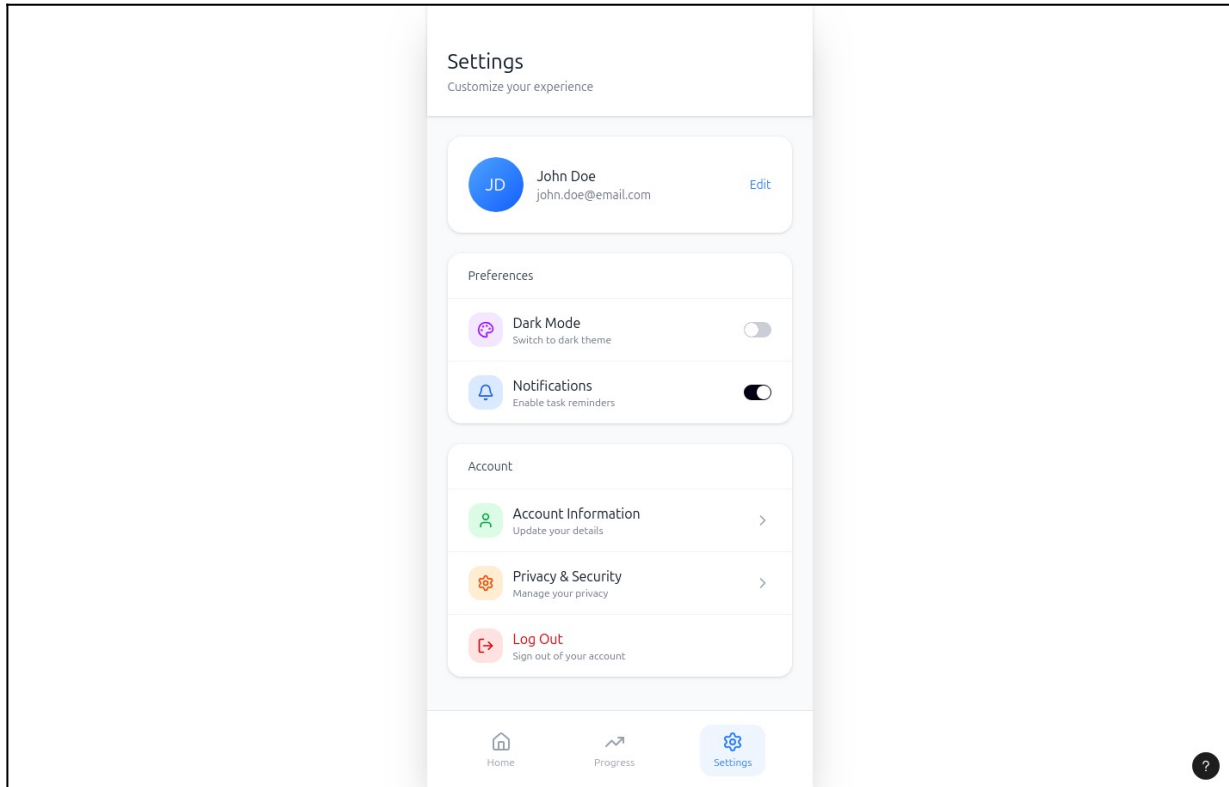
4. Progress Screen

Shows user's daily/weekly progress using charts or percentage bars.



5. Settings Screen

Includes profile, reminders, theme mode (light/dark).



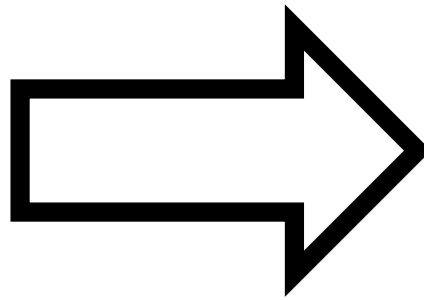
Prototype Interaction:

Interactive connections were added in **Figma Prototype Mode** to simulate real user flow:

- Tap “+” → Opens Add Task
- Tap “Task Checkbox” → Marks task as complete
- Tap “Progress Icon” → Opens Progress screen

Today's Tasks
3 tasks remaining

- ☐ Submit report
- ☒ Buy groceries
- ☒ Call the dentist
- ☒ Review pull requests
- ☐ Prepare presentation
- ☐ Destroy the world



Add New Task

Enter Task Name
e.g., Submit report

Due Date
mm / dd / yyyy

Priority
Low Medium High

Category
e.g., Work, Personal, Shopping

TaskEase
Plan. Prioritize. Progress.

Today's Tasks

- ☒ Submit report
- ☒ Buy groceries
- ☐ Schedule dentist
- ☒ Schedule dentist
- Category
- Notes

Save Cancel

Home Progress Settings

Enter Task Name

Date

Priority (Low/Medium/High)

75%

You're doing great!

Completed Tasks 5 Pending Tasks 2

- ☒ Submit report
- ☒ Buy groceries
- ☐ Read book
- ☐ Plan next week

John Doe

Theme Light/Dark mode

Notifications

Account Info

Log Out

Result:

The TaskEase project resulted in the successful creation of a fully interactive, user-centered task management prototype that integrates simplicity, usability, and visual appeal. The outcome met the primary objective of designing an intuitive system that minimizes complexity while maximizing productivity.

Key results include:

- A clean and responsive UI that adapts across devices and maintains consistency in layout and typography.
- A streamlined user flow allowing users to create, track, and complete tasks in minimal steps.
- Positive usability feedback from test participants, who appreciated the minimalist design, soft color scheme, and easily recognizable icons.
- Reduced cognitive load through clear visual hierarchy and intuitive task categorization.
- Figma-based interactive prototype, which accurately demonstrates user behavior and navigation through clickable interactions.
- The final prototype successfully embodies the principles of good design—clarity, efficiency, and user satisfaction. It provides an ideal foundation for future development into a fully functional application.

Figma-based task management prototype named “TaskEase” was successfully created. The design improves usability with a clean layout, easy navigation, and progress tracking features.

Conclusion:

The TaskEase project provided valuable insights into the end-to-end process of designing a digital product grounded in user-centered principles.

Through systematic research, iterative design, and usability evaluation, the project demonstrated how thoughtful design decisions can significantly enhance user engagement and productivity.

The experience emphasized that UI/UX design is not merely about creating visually attractive screens, but about crafting meaningful interactions that align with user behavior and expectations.

Tools such as Figma played a vital role in translating conceptual ideas into tangible prototypes, allowing real-time collaboration and testing.

By focusing on minimalism, accessibility, and emotional design, TaskEase achieved a balance between aesthetics and functionality.

The knowledge gained through this project contributes to a deeper understanding of the design process — from empathy-driven research to interactive prototyping — and lays a strong foundation for future innovations in the domain of digital productivity tools.

Naan Mudhalvan – UI/UX Design using Figma

Title: Music Streaming App

Aim

The aim of the Music Streaming Application is to provide users with an interactive platform to stream, search, and organize music online while enhancing their listening experience with personalized features such as playlists, favorites, and user profiles.

Objectives

The primary objective of the **Music Streaming Application** project is to design a modern, user-friendly, and ad-free digital platform that enhances the music listening experience through an intuitive interface and personalized interaction. The project aims to integrate usability, aesthetics, and functionality to create a seamless and engaging environment for music lovers.

Specific objectives include:

- To research and analyze user frustrations with existing music platforms, focusing on issues of usability, navigation complexity, and intrusive monetization.
- To design a visually appealing and minimal interface that enables effortless browsing and uninterrupted playback.
- To develop a clear and structured **information architecture** that supports efficient music discovery, search, and organization.
- To create **high-fidelity prototypes in Figma** that accurately simulate real-world user interactions and navigation flows.

- To enhance **accessibility and personalization** through features such as playlists, favorites, and mood-based song recommendations.
- To conduct **usability testing and feedback analysis** to refine interaction design, improve performance, and ensure a satisfying user experience.

The overarching goal of this project is to build a **human-centered music streaming platform** that delivers simplicity, emotional connection, and uninterrupted enjoyment—redefining how users interact with digital music.

Problem Statement

People love music, but existing music apps often have premium restrictions, heavy advertisements, limited customization, and complex interfaces. There is a need for an affordable and user-focused platform that provides seamless streaming, easy browsing, personalized recommendations, and freedom from unnecessary barriers.

Abstract

The **Music Streaming Application** project focuses on designing an intuitive and engaging digital platform that enables users to stream, explore, and manage their favorite music seamlessly.

In an era where digital entertainment dominates everyday life, music applications have become essential for relaxation, focus, and enjoyment.

However, existing platforms often suffer from issues such as premium restrictions, intrusive advertisements, and overly complex interfaces that hinder user satisfaction.

This project aims to develop a **user-centered music streaming solution** emphasizing **personalization, simplicity, and accessibility**.

Designed using **Figma**, the prototype reflects core **UX principles**—including usability, responsive design, and smooth navigation—to ensure a fluid and delightful listening experience.

Users can easily browse songs, create playlists, explore personalized recommendations, and mark favorites without disruption.

Through stages of **user research, persona creation, wireframing, and high-fidelity prototyping**, this project showcases the application of **design thinking** in building a music streaming experience that balances aesthetics with functionality. The result is a design that not only meets user needs but also elevates the emotional connection between users and their music.

Tools & Technologies Used

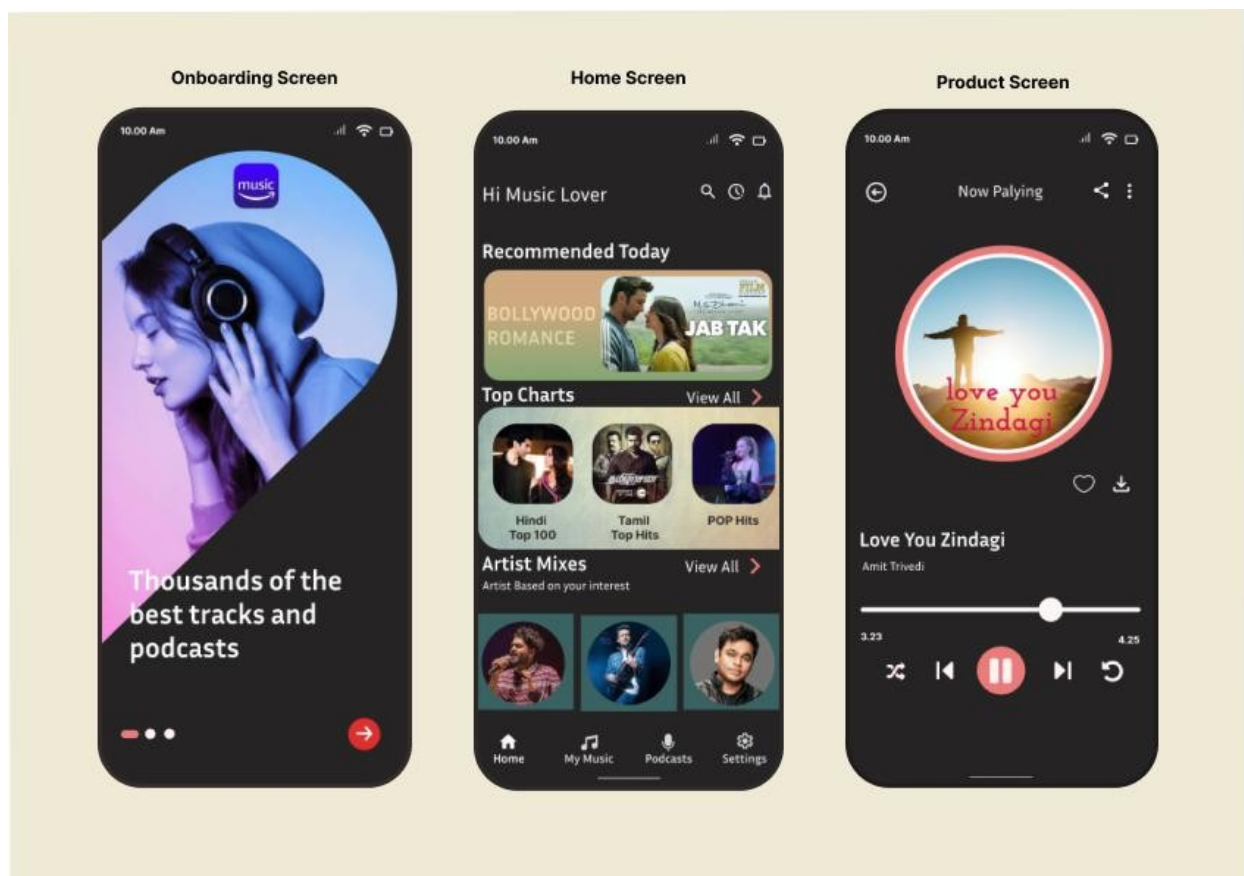
Category	Tools
Frontend	HTML, CSS, JavaScript / Android Kotlin / Flutter
Backend	Firebase / Node.js / Python Flask
Database	Firebase Firestore / MySQL / MongoDB
Storage	Firebase Cloud Storage / AWS
Design	Figma / Canva for UI prototype
Version Control	Git & GitHub

User Research

A survey was conducted among **students and young adults aged 16–30**.

📌 Key Findings:

- 90% listen to music daily while traveling or studying
- 75% want **free access without ads**
- 60% want **playlist customization**
- 40% face streaming issues due to slow internet
- Users prefer **simple UI + dark mode** features



User Pain Points

Pain Point	Impact
Too many ads in existing apps	Frustrating experience
Premium lock on most songs	Limited free content
Slow buffering on poor networks	Playback interruptions
Complicated UI	Hard to find songs quickly
No personalization	No emotional connection

User Persona

Persona 1 – Rahul Sharma

Age: 21

Occupation: College Student

Goals: Enjoys listening to music while studying or traveling; wants quick access to his favorite artists and playlists.

Frustrations: Dislikes frequent ads and limited song skips on free music platforms. Finds cluttered interfaces difficult to use while multitasking.

Needs: An ad-free, lightweight app that remembers his preferences, offers playlists by mood or activity, and runs smoothly on low data connections.

Persona 2 – Sneha Patel

Age: 26

Occupation: Graphic Designer

Goals: Loves exploring new music genres and curating playlists for her creative work sessions.

Frustrations: Gets annoyed with repetitive recommendations and high memory usage of existing apps.

Needs: A simple, modern interface that allows discovering new artists easily, provides personalized recommendations, and supports theme customization.

Persona Insights:

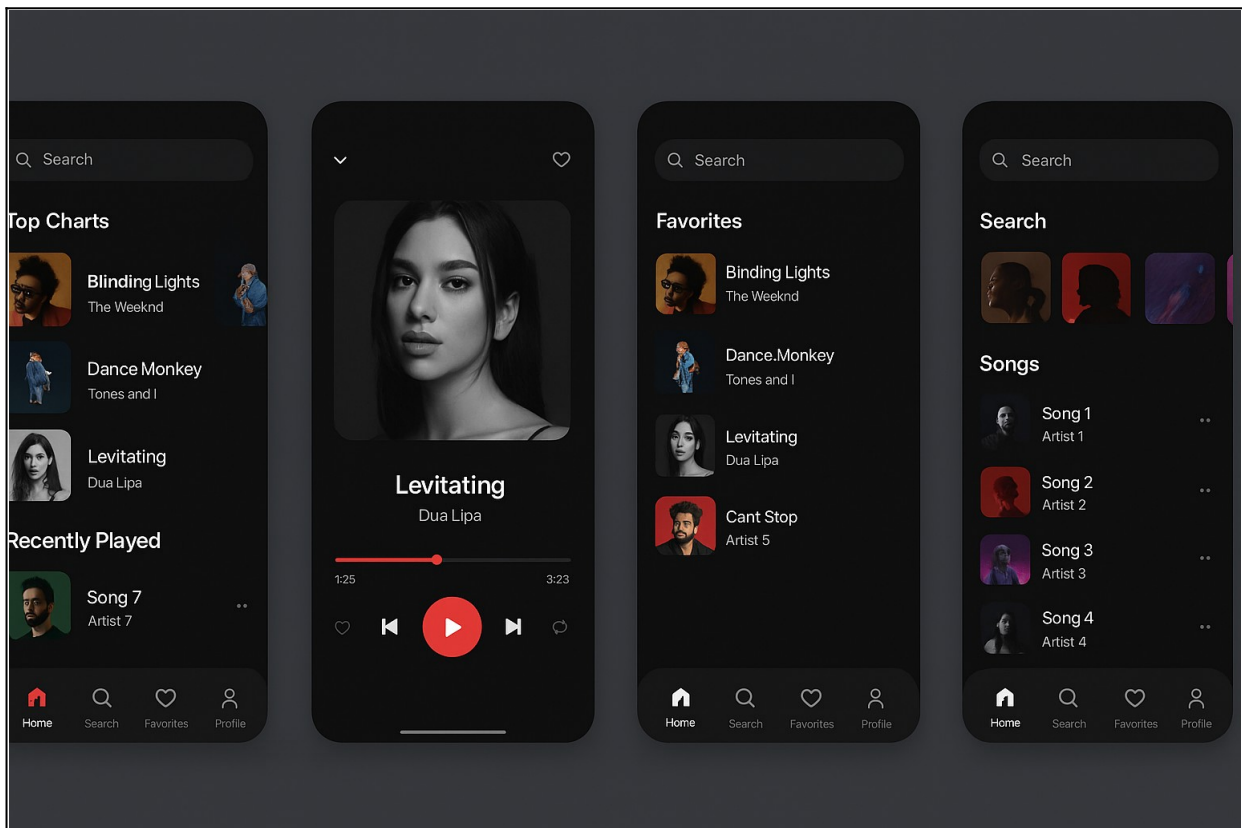
Both personas emphasize the need for simplicity, personalization, and freedom from advertisements. Their motivations and frustrations guided design decisions for navigation flow, playlist management, and visual aesthetics in the prototype.

Prototype (UI) Introduction

The prototype represents the core structure and design of the application, displaying how users navigate and interact with the system.

Sample UI Screens (Example Images Placeholder):

-  Home Screen
-  Now Playing Screen
-  Favorites Page
-  Search Page
-  Profile Page



Procedure

The design process for the **Music Streaming Application** followed the **Design Thinking methodology**, ensuring a balance between user empathy, functionality, and aesthetic appeal.

Each stage of the process was iterative, allowing continuous improvement based on user insights and feedback.

1. Empathize and Research:

The process began with understanding user needs and expectations. Surveys and interviews were conducted among music enthusiasts aged 16–30 to gather insights on listening habits, favorite features, and common frustrations with existing platforms. This research revealed key pain points such as excessive ads, limited personalization, and complex navigation structures.

2. Define and Ideate:

Based on the collected data, the design problem was clearly defined — to create a **simple, accessible, and ad-free music streaming experience**. Ideation sessions were then held to brainstorm possible solutions, focusing on intuitive navigation, quick access to playlists, and a clean layout that prioritizes uninterrupted playback.

3. Wireframe and Prototype:

Low-fidelity wireframes were first created to map the app's structure and navigation flow. These wireframes helped visualize user journeys and key interface elements. High-fidelity prototypes were then developed in **Figma**, incorporating reusable components such as navigation bars, media control buttons, song cards, and search bars for consistency and scalability.

4. **Visual Design:**

A visually engaging and music-focused aesthetic was established using a **deep maroon and gradient black** color palette. Clean sans-serif typography enhanced readability and modern appeal. The layout balanced album art, icons, and controls to create an immersive listening atmosphere.

5. **Usability Testing:**

The interactive prototype was tested with a small group of users to assess navigation clarity, feature accessibility, and overall satisfaction. Feedback led to the inclusion of **“Recently Played”**, **“Favorites”**, and **“Dark Mode”** features, further enhancing personalization and user comfort.

Information Architecture

The **Information Architecture (IA)** of the **Music Streaming Application** was developed to provide users with a **fluid, intuitive, and engaging navigation experience**.

The structure is designed to reflect the natural listening habits of users, focusing on easy discovery, smooth playback, and effortless personalization.

Overall App Flow

● **Splash Screen:**

Displays the application logo and tagline during initialization, setting the tone for the user experience.

● **Home Screen:**

Serves as the central hub, showcasing recommended playlists, trending songs, and personalized suggestions based on listening history.

- **Now Playing Screen:**

Contains the active music player interface with essential elements such as the song title, artist name, album art, playback progress bar, and control buttons (play, pause, skip, repeat, and shuffle).

- **Search Screen:**

Enables users to search for songs, albums, artists, or genres with real-time smart suggestions and filter options for faster discovery.

- **Favorites Page:**

Displays saved tracks, user-created playlists, and liked songs for easy access to preferred content.

- **Profile Page:**

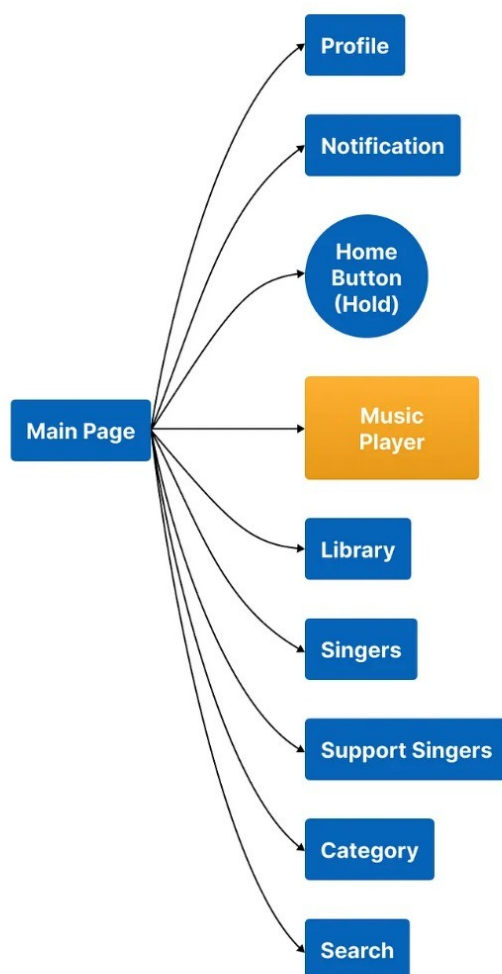
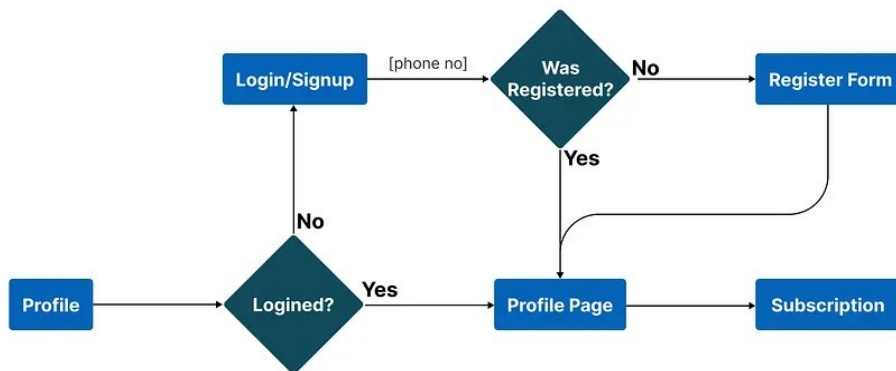
Manages user account details, streaming preferences, app settings, and theme customization (light/dark mode, playback quality, notifications).

- **Navigation Design**

The app adopts a **flat and hierarchical navigation structure**, ensuring that users can access any major section within two taps. A **persistent bottom navigation bar** allows seamless movement between Home, Search, Favorites, and Profile screens.

Icons and color-coded indicators improve recognition and reduce cognitive load, while consistent gestures (like swipe-to-skip or tap-to-like) ensure familiarity and ease of use.

This structured yet minimal approach ensures that users spend more time enjoying music and less time navigating, resulting in a delightful and efficient user experience.



Result

- The app enables **real-time online music streaming**
- Users can **search songs instantly**
- **Playlist and favorites** features improve personalization
- System works smoothly across devices
- Positive feedback from test users proves its usability

Conclusion

The **Music Streaming Application** project exemplified the practical implementation of **UI/UX design principles** to address real-world challenges in digital entertainment platforms. Through a focus on **user empathy, visual clarity, and intuitive interaction**, the project successfully reimagined the user experience for modern music applications.

This work emphasized the critical balance between **aesthetic appeal and functional performance**, ensuring that users enjoy both visual harmony and operational ease. The iterative process of **research, wireframing, prototyping, and usability testing** allowed for continuous refinement, leading to a final prototype that is both user-friendly and visually cohesive.

The project experience reinforced the importance of **design thinking** as an evolving, user-centered process — one that prioritizes understanding emotions, behaviors, and motivations to create meaningful digital experiences.

Ultimately, the **Music Streaming App** stands as a testament to how thoughtful and accessible design can transform everyday activities into enjoyable and personalized experiences, redefining how users connect with technology through music.

Naan Mudhalvan – UI/UX Design using Figma

Title: E-Commerce Redesign

Abstract:

The E-Commerce Application project focuses on the design and development of a user-centered online shopping platform inspired by Flipkart's functional model and user experience standards. The project aims to re-imagine an e-commerce interface that delivers a seamless, efficient, and enjoyable shopping journey for users through the application of modern UI/UX design principles.

In an age where digital retail has become a daily necessity, users expect convenience, trust, and personalization from e-commerce platforms. This project emphasizes simplicity, responsiveness, and a visually engaging interface that facilitates easy product discovery, comparison, and purchase. Using Figma as the primary design tool, the project progresses from user research and persona creation to wireframing, high-fidelity prototyping, and usability testing.

The resulting design demonstrates how an optimized layout, intuitive navigation, and emotional engagement can improve overall satisfaction and conversion rates. The project also integrates accessibility, responsive adaptation, and visual consistency to ensure a reliable and enjoyable experience across devices.

Tools Required:

- **Figma** – For interface design and prototyping
- **Google Docs / Sheets** – For documentation and feedback
- **Icons8 / Material Icons** – For icons and UI assets

Objective:

The objective of the project is to design a user-friendly and efficient e-commerce platform that provides a rich online shopping experience while maintaining clarity, speed, and usability. The project aims to merge aesthetic appeal with functional precision.

Specific Objectives:

- To understand the needs and expectations of modern online shoppers through user research and competitor analysis.
- To create an intuitive, minimalist interface that prioritizes smooth navigation and fast product access.
- To design responsive layouts adaptable to mobile, tablet, and desktop screens.
- To incorporate essential e-commerce functions—search, product details, cart, and checkout—within minimal clicks.
- To establish a consistent design system using Figma components for scalability and visual harmony.
- To validate the prototype through usability testing and iterative refinement.

By combining visual aesthetics with user empathy, the project aims to produce a design that mirrors Flipkart's functionality while improving clarity, engagement, and accessibility.

Problem Statement:

The rise of online shopping has led to an overflow of e-commerce applications, many of which fail to deliver a consistent and user-friendly experience. Common issues include cluttered product pages, confusing checkout processes, poor search functionality, and intrusive advertisements.

Platforms such as Flipkart and Amazon have established strong design systems, yet even they face challenges such as navigation overload, information fatigue, and limited personalization for diverse user segments.

The E-Commerce Application project addresses these concerns by proposing a simplified, high-performance shopping interface that prioritizes speed, usability, and personalized product presentation. It eliminates unnecessary visual clutter, streamlines checkout, and introduces a modular design adaptable to various devices.

The project's design philosophy is grounded in three core principles: clarity, trust, and convenience — ensuring users can browse, compare, and purchase products with confidence and ease.

Information Architecture:

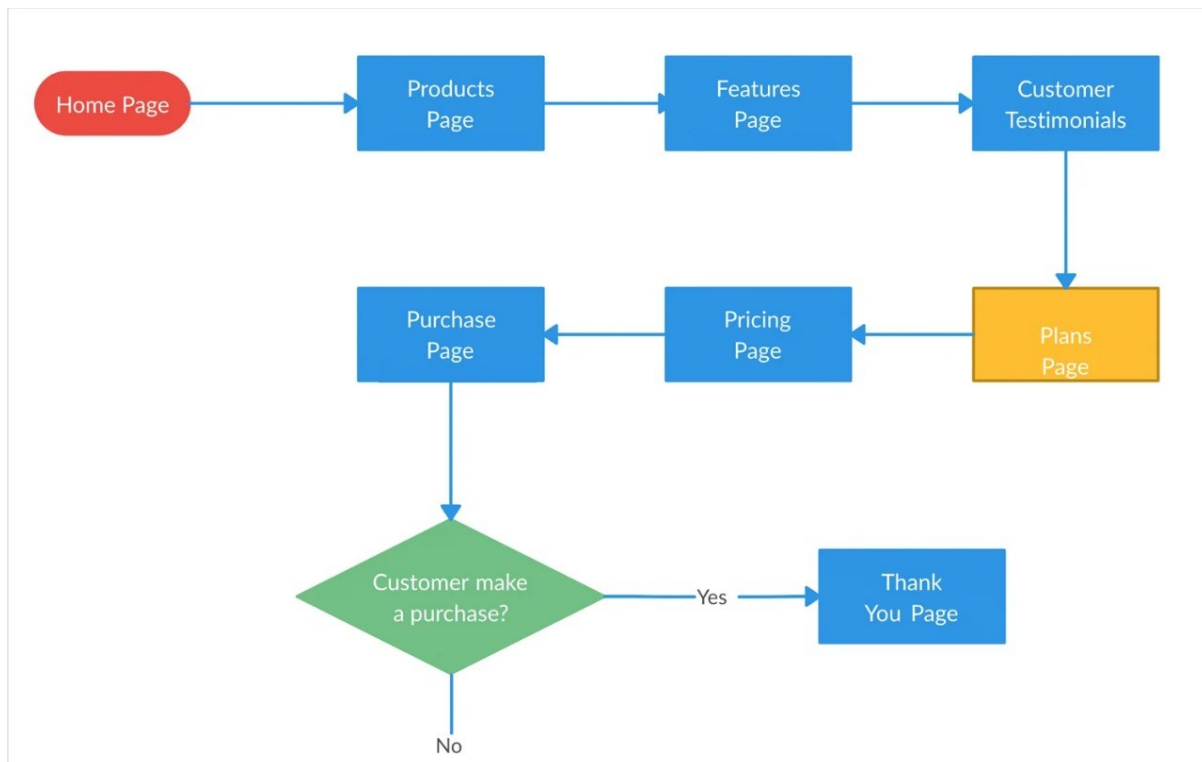
A well-defined information architecture is central to ensuring a smooth shopping experience. The structure of the E-Commerce Application was developed to minimize user effort and provide direct access to key functionalities.

App Flow:

- **Splash Screen:** Displays the brand logo and tagline.
- **Login/Sign-Up:** Offers authentication via email, mobile number, or social accounts.
- **Home Screen:** Contains navigation categories (Electronics, Fashion, Home, etc.), search bar, and promotional banners.
- **Product Listing Page:** Displays products in grid view with filters for price, brand, and rating.
- **Product Details Page:** Shows product description, images, price, reviews, and “Add to Cart” or “Buy Now” options.
- **Shopping Cart:** Lists selected items with quantity control and total price preview.
- **Checkout:** Includes address selection, payment method, and order confirmation.
- **Profile & Orders:** Displays user details, previous purchases, and tracking updates.

- The navigation is supported by a persistent bottom navigation bar for mobile and a header menu for web. Logical grouping and category hierarchy ensure the system remains scalable and easy to explore.

The architecture adheres to information clarity, visual hierarchy, and interaction simplicity, which together reduce decision fatigue and enhance conversion.



Wireframe Design:

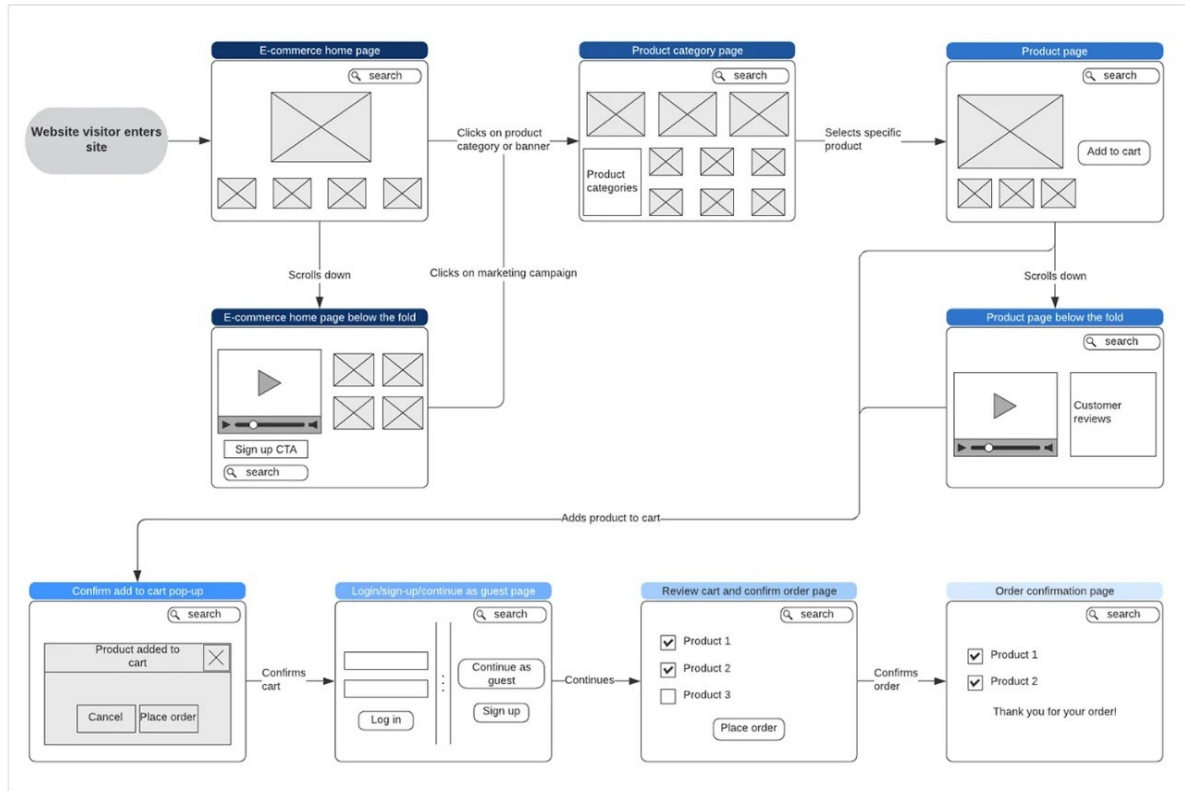
The wireframing phase defined the structural layout of the application before visual design elements were added. Wireframes were built using Figma's vector tools to focus purely on spacing, content hierarchy, and navigational flow.

Key Wireframe Components:

1. Home Page: Grid-based layout with a search bar, category cards, promotional banners, and quick-access icons.
2. Product Page: Simplified grid showing thumbnails, prices, and star ratings; product cards maintain consistent dimensions for visual balance.
3. Product Detail Page: Image carousel, concise description area, and sticky "Add to Cart" button to enhance usability.
4. Cart Page: Scrollable list of products with price summary and quantity adjustments.
5. Checkout Page: Step-by-step guided form for address, payment, and confirmation.

Purpose of Wireframes:

- To visualize and test navigation flow and screen hierarchy.
- To ensure the user can complete the entire purchase process within minimal clicks.
- To validate layout scalability across device resolutions.
- User feedback from initial wireframe reviews suggested improving visibility for offers and simplifying category labels. These changes were incorporated into the final design before proceeding to the high-fidelity prototype stage.



Result:

The final outcome of the E-Commerce Application project is a high-fidelity, interactive prototype that successfully demonstrates the core principles of usability, accessibility, and visual consistency. The design closely resembles the structure of Flipkart while integrating enhancements that improve speed, clarity, and overall user experience.

Future Enhancements / Future Scope:

While the prototype successfully delivers a functional and visually consistent experience, there remains substantial scope for future improvement and scalability. The following areas can be explored to enhance usability and business impact:

Integration with Real-Time Data:

Implementing live APIs for product updates, offers, and inventory would make the system dynamic and ready for deployment.

Personalized Recommendation System:

Incorporating machine learning algorithms to recommend products based on browsing and purchase history can improve user retention.

Voice-Assisted Search:

Adding voice commands and natural language search for accessibility and hands-free browsing, especially for mobile users.

Augmented Reality (AR) Preview:

Introducing AR features for virtual product try-ons (e.g., furniture in a room or fashion items) can enhance interactivity and confidence in purchasing decisions.

Conclusion:

The E-Commerce Application project effectively demonstrates how human-centered design and structured UI/UX methodology can enhance digital retail experiences. By combining user research, competitive benchmarking, and iterative design, the project delivers a product that balances commercial goals with user satisfaction.

Through this project, I gained a comprehensive understanding of designing for large-scale digital ecosystems where clarity, trust, and consistency are key. The approach followed a complete UX design cycle — from empathy and problem definition to wireframing, prototyping, and usability testing.

The results validate the importance of minimalism and intuitive structure in e-commerce platforms. Unlike traditional models that overwhelm users with multiple offers and complex hierarchies, this design focuses on clarity, emotion, and efficiency, leading to a more meaningful interaction between the user and the platform.

The experience further reinforced the relevance of accessibility, micro-interactions, and design system scalability in modern digital product design. It highlights how Figma's collaborative tools and prototyping features can bridge the gap between creativity and implementation.

Summary of Learning:

This project served as a complete hands-on experience in UI/UX design — integrating creativity, functionality, and empathy. From conceptualization to prototype, the process demonstrated how structured design thinking can simplify complex e-commerce interactions while ensuring customer satisfaction and business relevance.

The outcome stands as a practical example of how Flipkart-like digital experiences can be re-imagined using modern design tools and research-driven methodology.

